AI Work Portal

A NAS-native workspace for multimodal AI

Unify file intake, meeting recording, speech transcription, transcript translation, document OCR, vector retrieval, generative models, and complete auditability from one operational portal.

1

NAS data sovereignty

Keep source assets, indexes, vectors, and audits together

2

Local and cloud AI

Choose execution location and model per workload

3

Traceable automation

See discovery, processing state, source, and model end to end

All persistent information is stored on the NAS

Source files, transcripts, translations, document chunks, vectors, answer cache, model weights, and call records remain on NAS storage. Content leaves the NAS only when a user explicitly selects a cloud model for that operation.

CORE CAPABILITIES

24×7	8	2-tier	ZH / EN
NAS intake watch	Core AI workflows	Answer cache	Bilingual UI

A NAS-first data lifecycle

DATA SOVEREIGNTY / GOVERNANCE / LIFECYCLE

AI Work Portal brings shared storage, access control, AI processing, and knowledge search into one entry point. Every file is assigned a NAS asset ID that links its source, processing stages, model output, and audit trail.

01

Secure intake

Web upload, browser recording, SMB / NFS / WebDAV

02

Instant discovery

Notify users and enqueue each new file immediately

03

Content processing

ASR, translation, OCR, YOLO, and document parsing

04

NAS indexing

Chunks, pages, previews, vectors, and metadata

05

Retrieve and answer

Hybrid RAG, answer cache, and LLM execution

What stays on the NAS

Source assets

Audio, video, PDF, DOCX, and general files with filename, path, MIME type, size, uploader, and timestamp.

Derived content

Transcripts, translations, OCR text, video detections, rendered page images, and source metadata.

Knowledge index

RAG chunks, 1024-dimensional embeddings, page numbers, chunk types, and access-protected previews.

Operational records

Processing state, errors, model input and output, caller, token usage, timestamps, and answer-cache hits.

Access, versioning, and live status

- Login and session cookies protect asset APIs, recordings, and source-page images.
- Users see their own data; administrators manage the full asset scope with a clear integration point for NAS ACLs.
- AI output never overwrites source files, supporting reprocessing, version comparison, and NAS snapshots.
- WebSocket updates report intake, processing, completion, missing-model, and failure states in real time.

Automatic model discovery, download, and management

MODEL OPERATIONS / LOCAL AI / CLOUD CHOICE

The portal automatically reads the model catalog and local weight state, detecting runtime readiness, downloaded size, integrity, and availability. Downloads remain user-initiated so NAS capacity and network usage stay controlled.

1. **Discover** — Load local and cloud model catalogs
2. **Inspect** — Check runtime, weights, size, and integrity
3. **Download** — Show progress with cancel and resumable retry
4. **Ready** — Only complete models enter processing
5. **Audit** — Record every successful or failed call

MODEL PORTFOLIO

Speech / ASR

whisper.cpp small / base, faster-whisper, SenseVoiceSmall, and Paraformer-zh; cloud options include OpenAI Transcribe, Deepgram, and AssemblyAI.

Translation and generation

Local Qwen3-4B-Q4_K_M on the NAS; cloud choices include OpenAI, Claude, Gemini, DeepSeek, Qwen, Mistral, Groq, and more.

Embeddings

Qwen3-Embedding-0.6B runs on CPU and creates 1024-dimensional vectors for documents and queries.

Vision and OCR

YOLO frame sampling and object detection; PyMuPDF page rendering with PaddleOCR for image-based PDFs.

NAS Model Management panel

Installation state

Installed, partial, missing, downloading, cancelled, or failed

Capacity details

Expected size, downloaded bytes, model path, and progress percentage

Operator controls

Download, cancel, retry, and refresh with partial-file resume support

Runtime isolation

Separate dependency and device management for web, generation, embedding, ASR, and vision

Multimodal processing, Hybrid RAG, and specifications

MULTIMODAL / HYBRID RAG / AUDIT

Meetings and audio

Preserve source voice; choose local or cloud ASR; then choose a target language and translation LLM; index source and translated text separately.

PDF / DOCX

Use pypdf for text PDFs; render image PDFs with PyMuPDF and read them with PaddleOCR; show page citations and previews with answers.

Video

Sample frames with YOLO and store objects, confidence, timestamps, and previews as video_detection chunks.

AI Work

Summarize, query, and analyze NAS assets with local Qwen or mainstream cloud models, including pricing and API-key requirements.

Hybrid RAG and cost control

75%	25%	0.76	2-tier
Vector similarity	**Keyword score**	**Semantic cache threshold**	**Answer cache**
Qwen3 Embedding + cosine similarity	Multilingual term matching	Reuse a relevant answer already held on the NAS	Normalized exact match + semantic match

Reference deployment

Web / API	FastAPI, HTTP 80 redirected to HTTPS 443, session login, and role-based access
Persistence	NAS filesystem + SQLite for assets, meetings, chunks, vectors, cache, and call audits
Local generation	Qwen3-4B-Q4_K_M / llama.cpp, OpenAI-compatible API, 4096-token context
Embedding	Qwen3-Embedding-0.6B, CPU, 1024 dimensions, independent service
Service control	systemd-managed portal and model services with startup and restart policies
Observability	Processing timeline, model state, errors, tokens, balance, and complete input/output records